

Insights Report

Danish Pharmacy Sales Analysis

Roskilde, Denmark • April 2023

Executive Summary

This report presents findings from the analysis of Danish Pharmacy Sales Data covering 1–13 April 2023. The dataset comprises 20,003 transactions across 100 medicines. Analyses span daily revenue and quantity trends, hourly and weekday sales patterns, top-selling medicine rankings by both revenue and volume, and drug category distributions. Key findings indicate stable overall demand with identifiable peak days and hours, a strong positive correlation between quantity sold and revenue, and dominant revenue contributions from Psychotropic and Immunosuppressant drug classes.

Sales Trend Analysis

1. Daily Revenue Trend

Observed Trends

- Overall revenue was broadly stable, ranging between ₹950,000 and ₹1,050,000 per day.
- Revenue peaked on April 6 (highest) and April 11 (second highest).
- A sharp decline occurred on April 13, with revenue falling significantly below the daily average.

Recommendations

- Ensure adequate stock and staffing on historically high-revenue days (April 6 and April 11 pattern).
- Investigate the April 13 anomaly — determine whether it reflects a supply issue, reduced footfall, or a data recording gap.

Limitations

- Trend reflects revenue only; units sold may tell a different story.
- External factors such as holidays, promotions, and doctor availability are not captured.

2. Daily Quantity Trend

Observed Trends

- Daily units sold remained stable between 8,400 and 9,100 units across most of the period.
- Quantity peaks on April 6 and April 11 align with the revenue peaks, confirming demand-driven spikes.
- April 13 recorded the lowest quantity at below 5,000 units — consistent with the revenue anomaly.

Recommendations

- Use stable daily quantity figures to anchor weekly stock order planning.
- Ensure staffing and inventory buffers are in place ahead of historically high-demand days.

Limitations

- Quantity trend provides no insight into revenue or margin contribution.
- No segmentation by drug category or medicine type is included.

3. Correlation: Revenue vs Quantity

Observed Trends

- A strong positive correlation exists between daily units sold and daily revenue.
- Most days cluster tightly between 8,400–9,100 units and ₹950,000+ in revenue.
- April 13 is a clear outlier, sitting significantly below the cluster on both dimensions.

Recommendations

- Initiatives that increase sales volume (footfall, repeat purchases, prescription fulfilment) are likely to translate directly into revenue growth.

Limitations

- External factors affecting both metrics simultaneously — such as supplier delays or local events — are not accounted for.

4. Hourly Sales Pattern

Observed Trends

- Sales begin rising from 10 AM, peak between 11 AM and 1 PM, then gradually decline through the evening.
- A stable plateau is observed from 3 PM to 4 PM before the evening decline resumes.

Recommendations

- Schedule peak staffing between 10 AM and 2 PM to meet demand without service degradation.
- Use off-peak hours (early morning, late afternoon) for restocking and administrative tasks.

Limitations

- The chart reflects total revenue, not transaction count or customer footfall.
- Weekend and holiday patterns may deviate from weekday norms.

5. Day-Hour Sales Heatmap

Observed Trends

- Monday through Wednesday between 10 AM and 2 PM consistently show the highest revenue concentration.
- Friday records the lowest sales activity, particularly after 2 PM.
- Notable weekend spikes occur at Saturday 3 PM and Sunday 4 PM.

Recommendations

- Concentrate staffing and operational capacity on Monday–Wednesday mornings.
- Consider reduced staffing on Friday afternoons and reallocating those hours to Saturday PM coverage.
- Stock replenishment schedules should account for the Monday–Wednesday demand surge.

Limitations

- Revenue concentration may reflect price rather than customer volume.
 - Public holidays, local events, and seasonal factors are not reflected in this view.
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Medicine-Level Analysis

6. Top 10 Medicines by Revenue

Observed Trends

- Azithromycin, Saxagliptin, and Sertraline are the top three revenue contributors.
- Clonazepam and Clindamycin also feature prominently, likely driven by a combination of volume and unit price.

Recommendations

- Maintain close monitoring of stock levels and expiry dates for top-earning medicines.
- Explore bulk purchasing or preferential supplier terms for high-revenue drugs to protect margins.

Limitations

- Revenue-based ranking may overrepresent high-priced medicines sold infrequently.
- Units sold is not reflected — a medicine may rank highly here but move slowly.

7. Top 10 Medicines by Quantity Sold

Observed Trends

- Escitalopram, Senna, and Oxycodone record the highest unit volumes, indicating strong and consistent demand.
- Azithromycin and Saxagliptin appear in both the revenue and quantity top 10, marking them as dual-priority medicines.

Recommendations

- Ensure reliable and continuous stock availability for high-volume medicines to prevent shortages.
- Allocate shelf space and storage proportional to each medicine's turnover velocity.

Limitations

- Quantity rank does not reflect profitability — low-cost medicines may dominate by volume while contributing marginally to revenue.
- Expiry data is not considered; this is especially relevant for high-volume bulk medicines.

8. Revenue vs Quantity for Top 10 Earners

Observed Trends

- A positive correlation between quantity and revenue holds within the top 10 revenue medicines.
- Azithromycin leads in revenue; Clonazepam leads in quantity within this group.
- Saxagliptin and Sertraline perform strongly on both dimensions.
- Travoprost, Cetirizine, and Sitagliptin rank low on both metrics within the top 10.

Recommendations

- Prioritise Clonazepam stock depth — high demand warrants bulk purchasing consideration.
- Promote Azithromycin and Saxagliptin where appropriate — high revenue per unit sold.
- Review restocking frequency for low dual-metric performers to avoid overstocking.

Limitations

- Does not account for discounts, returns, or expiry-related write-offs.
 - The top-10 focus may obscure emerging high-potential medicines outside this group.
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Category-Level Analysis

9. Revenue Share by Drug Category

Observed Trends

- Psychotropic and Immunosuppressant categories contribute the largest share of total revenue.
- Antihistamine, Antifungal, and Gastrointestinal categories contribute the least to revenue.

Recommendations

- Prioritise stocking, expiry monitoring, and supplier negotiations for Psychotropic and Immunosuppressant drugs.
- Assess whether low-revenue categories reflect low demand or simply lower unit prices before reducing their shelf presence.

Limitations

- Revenue share does not reflect units sold or prescription frequency.
- Category performance may be influenced by seasonal factors and local prescription patterns not captured in the dataset.

10. Quantity Share by Drug Category

Observed Trends

- Immunosuppressant and Psychotropic categories dominate by units sold, consistent with their revenue leadership.
- Antifungal, Antihistamine, and Antibiotic classes contribute the least by volume.
- The distribution suggests that long-term treatment regimens are primary demand drivers.

Recommendations

- Negotiate volume-based supplier deals for Immunosuppressant and Psychotropic categories to protect margins on high-turnover stock.
- Investigate low-volume categories to determine whether limited demand reflects unmet need or adequate supply.

Limitations

- Quantity share does not reflect profitability or revenue contribution.
- Seasonal demand shifts and regional prescription trends are not included in this view.